

Campus Student Job System

System Request, Feasibility Analysis, and Project Plan

Deliverable 1 - System Request

IS 202 - Structured System Analysis and Methods

Team Name: Group 6

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1. Team Information

Item	Detail
Team name	Group 6
Project contact person	Harsh Danayak ~ harshd2@umbc.edu
Time set aside to meet outside class	Thursdays: 5:00-6:00 PM, plus Sunday check-in via Blackboard Messages.

Member	Contact	Short bio	Project role(s)
Harsh Danayak	harshd2@umbc.edu 301-401-7572	Majoring in Business Technology Administration	Project Manager / Scrum Master
Abhi Chandu	achandu1@umbc.edu 443-787-8231	Majoring in Business Technology Administration.	Systems Analyst
Shruti Gomes	sgomes3@umbc.edu 240-485-8832	Majoring in Business Technology Administration	Database Designer / Developer
Chandana Malla	cmalla2@umbc.edu 667-770-4018	Majoring in Business Technology Administration	UI/UX Designer / Tester
Cheta Greg Obi	ngregob1@umbc.edu 240-877-8173	Majoring in Business Technology Administration	Database Designer / Developer

2. Project Summary

Problem. Students who want on-campus jobs (library, dining, labs, recreation, administrative offices) find them through scattered channels: department bulletin boards, emails, word of mouth, and general-purpose platforms such as Handshake that are built mostly for off-campus internships and full-time roles. Departments post jobs in inconsistent ways, receive applications by email, and track candidates in spreadsheets. Students miss openings, and supervisors spend hours screening and coordinating.

Proposed solution. The Campus Student Job System (CSJS) is a web-based platform dedicated only to on-campus student employment. Departments post positions, students create profiles and apply with one click, supervisors review and schedule interviews, and the University Career Center gets a central view of postings, applicants, and hiring outcomes.

How the project fits the characteristics of a successful IT project

- **Unique purpose:** one place to find, post, and manage campus-only student jobs, tailored to work-study rules, class schedules, and hour caps.
- **Progressive elaboration:** we move from this general request to context-level and detailed data flow diagrams, an entity-relationship model, then design and build.
- **Resources from various areas:** Career Center, Financial Aid (work-study eligibility), Human Resources/Payroll, IT Services, and hiring departments.
- **Primary customer/sponsor:** the University Career Center.
- **Uncertainty and challenges:** integration with the student information system and single sign-on, privacy of student data, and adoption by departments that already use their own methods.

- **Estimates and cost:** see **Sections 5 and 7** for the three-year cost-benefit analysis and the dated work plan.

3. System Request

Element	Description
Project name	Campus Student Job System (CSJS)
Project sponsor	Director of the University Career Center (primary client). Supporting stakeholders: Financial Aid (work-study), Human Resources/Payroll, IT Services, department hiring supervisors, and student employees.
Business need	Students and campus departments lack a single, reliable channel for on-campus jobs. Postings are fragmented, hiring is slow and manual, and the Career Center has no consolidated data on student employment. Existing tools are designed for off-campus employers and do not handle campus-specific rules.
Business requirements (what the system will do)	- Student accounts with profile, resume upload, availability, and major/skills - Department accounts to post, edit, close, and renew positions - Search and filter by department, pay, hours per week, work-study eligibility, and schedule fit - One-click applications, status tracking, and email notifications - Supervisor tools: review applicants, shortlist, schedule interviews, record hiring decisions - Career Center dashboard and reports (postings, applications, hires, time-to-fill) - Role-based access and secure login through the university's single sign-on
Business value - tangible	Less administrative time per posting; lower printing and duplicate-posting costs; reduced vacancy time and rehiring cost for departments; faster work-study reporting. Estimated at about \$162,000 of benefits over three years (Section 5).
Business value - intangible	Better student experience and access to jobs, higher student engagement and retention, more equitable visibility of openings, stronger data for Career Center decisions, and improved campus employer brand.
Special issues and constraints	- Student data privacy (FERPA) and accessibility (WCAG) compliance - Must respect work-study eligibility and weekly hour limits - Integration with campus single sign-on and student information system may need IT Services approval - Fixed semester timeline; team is student-based with limited hours per week - Requires department buy-in and light training

4. Feasibility Analysis

Technical feasibility - High. CSJS uses mature, well-documented technology: a web front end, a relational database, and a cloud host. The team has coursework in database systems and networking. Main technical risk: single sign-on and student-record integration. Mitigation: start with standalone accounts (email verification) and add single sign-on as a later sprint.

Economic feasibility - High. Estimated three-year cost is about \$99,950 against estimated benefits of about \$162,000, for a net benefit of about \$62,050, an ROI of 62%, an NPV of about \$48,126 at an 8% discount rate, and break-even in roughly 20 months (Section 5). Benefits depend on adoption, so the plan assumes gradual growth from about 50% to 100% of campus postings.

Operational (organizational) feasibility - Medium to High. The Career Center sponsors the project and students are motivated users because the system saves them search time. The main risk is departments that prefer email or paper. Mitigation: involve two or three pilot departments early, keep the posting form under five minutes, and provide a one-page guide.

Schedule feasibility - Medium. A working prototype in one semester is realistic if scope is limited to a minimum viable product (post, search, apply, review, report). Integrations and advanced analytics are placed in later releases.

Legal and ethical feasibility - Medium. The system stores student information, so it must minimize data collected, restrict access by role, and follow FERPA and accessibility standards. This needs review by the university's IT and compliance offices.

Risk summary

Risk	Likelihood	Impact	Mitigation
Low department adoption	Medium	High	Pilot departments, simple forms, Career Center endorsement
Integration delays (SSO/SIS)	High	Medium	Standalone login first, integrate later
Scope creep	Medium	Medium	Prioritized backlog, MVP scope, sponsor sign-off each sprint
Student data privacy issue	Low	High	Role-based access, data minimization, compliance review

5. Cost-Benefit Analysis (Three Years, Starting 2024)

Item	Year 1 (2024)	Year 2 (2025)	Year 3 (2026)	Total
COSTS				
One-time development and launch	\$51,550	\$0	\$0	\$51,550
Cloud hosting	\$3,600	\$3,600	\$3,600	\$10,800
Email/notification service	\$1,200	\$1,200	\$1,200	\$3,600
Maintenance and support	\$6,000	\$12,000	\$12,000	\$30,000
Security and accessibility review	\$0	\$2,000	\$2,000	\$4,000
Total costs	\$62,350	\$18,800	\$18,800	\$99,950
BENEFITS				
Staff time saved on posting/screening (1,800 hrs at full adoption x \$25)	\$22,500	\$40,500	\$45,000	\$108,000
Reduced printing and duplicate postings	\$1,500	\$1,500	\$1,500	\$4,500
Lower vacancy and rehiring cost	\$5,000	\$10,000	\$12,000	\$27,000
Faster work-study/compliance reporting	\$4,500	\$9,000	\$9,000	\$22,500
Total benefits	\$33,500	\$61,000	\$67,500	\$162,000
Net benefit	(\$28,850)	\$42,200	\$48,700	\$62,050
Cumulative net	(\$28,850)	\$13,350	\$62,050	
Discount factor (8%)	0.9259	0.8573	0.7938	
Present value of net	(\$26,713)	\$36,180	\$38,660	\$48,126

Metric	Result
Total 3-year cost	\$99,950
Total 3-year benefit	\$162,000
Net benefit	\$62,050
ROI = net benefit / total cost	62.1%
NPV at 8%	\$48,126
Break-even point	about 20 months (during Year 2)

Key assumptions. These are planning estimates for a mid-sized campus (about 600 student job postings per year). Development labor is estimated at market rates (about 920 hours: analyst/PM 250 hrs at \$45, developers 400 hrs at \$35 for two developers, designer 120 hrs at \$40, QA 150 hrs at \$30) even though the class team is unpaid, so the sponsor sees the true cost. Adoption is assumed at 50%, 90%, and 100% of postings in Years 1-3. Staff time is valued at \$25/hour and compliance time at \$30/hour. Figures will be refined with real data from the Career Center during analysis.

6. Project Methodology: Agile (Scrum)

We will use **Agile with Scrum**, working in two-week sprints, with a prioritized product backlog managed on our GitHub Project board.

- **Requirements will evolve.** Departments and students will react to prototypes and change what they need; Agile welcomes this, while a strict waterfall approach would lock requirements too early.
- **Early, working results.** Building the system one subsystem at a time (accounts, postings, applications, reports) gives the sponsor something to review every sprint.
- **Fits a student team and fixed semester.** Short sprints, a visible board, and weekly stand-ups keep a small team on track and expose problems quickly.
- **Risk reduction.** Uncertain areas (adoption, integration) are tested early through pilots and prototypes.
- **Alignment with SDLC.** Each sprint still covers planning, analysis, design, implementation, and testing in miniature, and we will keep the documentation (DFDs, ERD, requirements) that this course requires.

Team practices

- Sprint planning at the start of each sprint, weekly stand-up, sprint review with the sponsor, and a sprint retrospective.
- Roles: Product Owner (represents the Career Center), Scrum Master (team member), and Development Team.
- Definition of done: feature built, tested, documented, and merged on GitHub.

7. Project Work Plan (Estimated Dates)

GitHub Project board link: <https://github.com/Sgomes3/IS-202-Group-6-Project>

Sprint / Milestone	Dates (est.)	Key tasks	Deliverable
Sprint 0 - Initiation	Sep 28 - Oct 9, 2026	Submit and refine system request; meet sponsor; set up repo and board; identify stakeholders	Approved system request
Sprint 1 - Analysis	Oct 12 - Oct 23, 2026	Interviews and surveys with students and departments; requirements list; context-level and Level-0 DFDs	Requirements document, DFDs
Sprint 2 - Design	Oct 26 - Nov 6, 2026	ERD and data dictionary; screen mockups; database schema; security and role design	ERD, prototype screens
Sprint 3 - Build core	Nov 9 - Nov 20, 2026	Accounts, job postings, search/filter, application submission	Working core release
Sprint 4 - Build and test	Nov 23 - Dec 4, 2026	Supervisor review tools, notifications, reports; testing; pilot feedback	Tested release candidate
Sprint 5 - Deploy and present	Dec 7 - Dec 11, 2026	Fixes, user guide, final documentation, final presentation	Final system and report

Dates should be adjusted to match the course calendar and instructor deadlines for later deliverables.

GitHub Project board setup

- **Columns:** Backlog, Ready, In Progress, In Review, Done.
- **Custom fields:** Sprint, Priority (High/Medium/Low), Estimate (story points or hours), Start date, Target date, Owner, Deliverable.
- **Issues:** one per task above (about 30-40), each assigned, estimated, and labeled (analysis, design, build, test, documentation).
- **Milestones:** one per sprint with the dates shown in the table.
- **Views:** Board (by status), Table (all fields), and Roadmap (timeline by target date).